Project Documentation

Project Title: EVENTFORCE MANAGEMENT SYSTEM

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EVENTFORCE MANAGEMENT SYSTEM

User Story

EventForce Management System is designed as a centralized Salesforce-based event management solution. It enables organizers to create and manage events, participants to register for events, and staff to monitor event activities through an intuitive interface supported by conversational AI. The system manages Event, Organizer, Attendee, Registration, and Venue information through Salesforce custom objects and relationships. Salesforce Flows automate event operations and notifications, while Agentforce AI and Prompt Builder allow users to retrieve event information and interact conversationally. Event staff can update event status and registration information in real time, while Experience Cloud, security controls, reports, and dashboards provide secure user access and operational visibility.

1. INTRODUCTION

EventForce Management System is a Salesforce-based event management solution developed to simplify event creation, registration, scheduling, and event management. The system supports the complete event lifecycle, from event creation and publishing to registration, event execution, and completion. It enables organizers to manage event information, while attendees can register and access event details. The solution uses Salesforce custom objects, relationships, Flows, Agentforce AI, Prompt Builder, Experience Cloud, reports, dashboards, and security controls to provide centralized and efficient event management.

1.1 Project Overview

EventForce Management System is a Salesforce CRM solution developed to simplify event planning, registration, scheduling, and event management. It supports the complete event lifecycle, from event creation and publishing through participant registration, event execution, and completion.

Organizers can manage event information, while attendees can register for events and access event details. Event staff can update event status and registration information.

1.2 Purpose

The purpose of EventForce Management System is to:

- Centralize event and participant information.
- Simplify event creation and registration.
- Provide real-time event visibility.
- Automate event updates and participant notifications.

- Enable conversational event assistance through Agentforce AI and Prompt Builder.
- Improve organizer efficiency through Salesforce Flows and dashboards.
- Provide secure, role-based access for organizers, attendees, support staff, and administrators.

2. IDEATION PHASE

2.1 Problem Statement

PS-1 Organizer: Wants to create and manage events easily, but event operations may require multiple applications or manual coordination because event information is not centralized. This can cause delays and confusion.

PS-2 Attendee: Wants to register for an event and access event information, but event details may be difficult to find or track. This can cause confusion.

PS-3 Event Staff: Needs to update event status and participant information, but manual updates can be time-consuming. Centralized event management is therefore needed.

PS-4 Attendee: Needs to receive event updates, but notifications may not be available at every event stage. Automated communication is required.

PS-5 Admin: Needs to monitor event operations, but event participation and performance can be difficult to monitor without centralized operational reporting.

2.2 Empathy Map Canvas

User Persona: Attendee / Participant

- **Think & Feel:** Wants a simple way to discover, register for, and participate in events without repeatedly contacting organizers.
- **Hear:** Receives event information, registration confirmations, reminders, and status updates.
- **See:** Event details, event date, venue, available seats, registration status, and event schedule.
- **Say & Do:** Registers for events, checks event information, and asks for event updates.
- **Pain:** Missing information, registration confusion, schedule changes, and dependence on organizers.
- **Gain:** Easy registration, real-time event information, convenient access, and timely notifications.

2.3 Brainstorming & Idea Prioritization

Ideas considered:

- Manual event registers
- Spreadsheet-based event tracking
- Standalone event portal
- Salesforce CRM with Agentforce AI

Final Selection

Salesforce CRM was selected because it provides centralized event management, automation through Flows, real-time event visibility, conversational AI assistance, secure role-based access, and reports and dashboards.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The participant journey consists of:

1. Event Discovery
2. Event Registration
3. Registration Confirmation
4. Event Participation
5. Event Completion
6. Feedback / Follow-up

3.2 Solution Requirements

Functional Requirements

- **FR-1** ➡ Event creation and record management
- **FR-2** ➡ Event status tracking
- **FR-3** ➡ Event capacity management
- **FR-4** ➡ Event date and schedule management
- **FR-5** ➡ Organizer information management
- **FR-6** ➡ Attendee information management
- **FR-7** ➡ Venue and event-location management

FR-8 Registration and attendance status updates

FR-9 Automated participant notifications

FR-10 AI-powered event information assistance

FR-11 Reports and dashboards

FR-12 Customer/participant access through Experience Cloud

Non-Functional Requirements

- **Usability** – Intuitive Salesforce and participant-facing interface
- **Security** – Role-based and field-level access control
- **Reliability** – Accurate event, registration, and participant information
- **Performance** – Timely Flow and Agentforce responses
- **Availability** – Salesforce cloud-based access
- **Scalability** – Supports growing event and participant operations
- **Data Integrity** – Controlled access and relationship-based event data
- **Maintainability** – Flows, prompts, topics, and configurations can be updated

3.3 Technology Stack

- **Platform:** Salesforce Developer Edition
- **UI:** Lightning App Page, Lightning Record Pages, Experience Cloud
- **Database:** Salesforce Custom & Standard Objects
- **Business Logic:** Salesforce Flows, Apex Classes
- **Automation:** Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
- **AI:** Agentforce AI, Prompt Builder
- **Notifications:** Email Alerts & Email Templates

- **Processing:** Batch Apex
- **Security:** Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
- **Reporting:** Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The EventForce Management System addresses common event-management challenges by providing a centralized Salesforce platform for event creation, registration, attendee management, scheduling, notifications, and reporting.

4.2 Proposed Solution

The proposed solution uses Salesforce CRM to centralize event information and automate event-management activities.

The system provides:

- Event creation and management
- Organizer management
- Attendee management
- Event registration
- Venue management
- Automated notifications
- Event status tracking
- Agentforce conversational assistance
- Prompt Builder AI responses
- Reports and dashboards

4.3 Solution Architecture

The solution architecture consists of:

Users → Experience Cloud / Lightning Interface → Salesforce CRM → Custom Objects → Flows / Apex → Agentforce / Prompt Builder → Reports & Dashboards

The main custom objects are:

- Event__c
- Organizer__c
- Attendee__c
- Registration__c
- Venue__c

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project follows an Agile methodology using Epics, User Stories, Story Points, sprint-based execution, and velocity-based estimation.

6. PROJECT DEVELOPMENT PHASE

Phase 1: Requirement Analysis & Planning

Objective

Event organizers and participants can experience delays and confusion while creating, registering for, and managing events when they need to use multiple

applications or contact support. A conversational AI agent can simplify the process by allowing users to obtain event information, check registration details, and manage event-related activities in one place.

Key Requirements

- Gather requirements from end users, organizers, and event staff.
- Define system scope and user personas: Attendee, Organizer, Staff, and Admin.
- Design the data model and relationships among Event, Organizer, Attendee, Registration, and Venue.
- Identify use cases such as event creation, registration, scheduling, and event-update automation.
- Select Salesforce Flows, Prompt Builder, and Agentforce AI.

Defining Project Scope & Objectives

Project Scope

1. Build a scalable CRM solution for event organizers to manage events, participants, registrations, and event operations efficiently.
2. Support the complete event lifecycle from event creation and publishing to registration, execution, and completion.
3. Include data model design, automation through Flows, Apex and Email Alerts, AI-powered Prompt Builder, and reporting dashboards.
4. Provide role-based access for administrators, event staff, organizers, and participants through Salesforce profiles, permission sets, and Experience Cloud.

5. Provide real-time event visibility for participants and simplified operational tracking for organizers through AI-integrated conversational support.

Objectives

- Streamline event operations through automated event creation, registration, and notifications.
- Improve participant experience through transparent event information and self-service registration.
- Enhance organizer efficiency through centralized dashboards and guided Flows.
- Provide business insights through reports and dashboards.
- Maintain security and reliability using role hierarchies, sharing rules, and controlled data access.

Gathering & Analysing User Needs

Users Involved

- **Attendees / Participants** – Register for events, view event details, and receive updates through the portal or AI agent.
- **Event Organizers** – Create and manage events, monitor registrations, and update event status.
- **Event Support Team** – Handle participant questions and event-related issues.
- **Admin Team** – Configure Salesforce, manage user roles and security, and monitor performance.
- **System Integrator / Developer** – Implement automation, AI agent configuration, and backend integrations.

Key Functional Needs

- Seamless event creation and registration through Salesforce Experience Cloud or agent assistance.
- Real-time event updates through stages such as:

Draft → Published → Registration Open → Registration Closed → Event Ongoing → Completed

- Automated notifications and email alerts for registration and event-stage updates.
- Organizer and attendee information management.
- Venue and schedule management.
- AI Agent integration through Prompt Builder.
- Reports and dashboards for event participation, registration, and organizer activity.
- Security controls using roles, profiles, and permission sets.

Tools Used

- **Salesforce Developer Org** – Configuration, development, and testing
- **Prompt Builder / Agentforce AI** – Intelligent event-management prompts and responses
- **Salesforce Flow Builder** – Event creation, registration, and notification automation
- **Google Sheets / Docs** – Initial data planning, documentation, and system mapping

Identifying Key Salesforce Features & Tools Required

Salesforce Features Planned

Custom Objects:

- Event__c
- Organizer__c
- Attendee__c
- Registration__c
- Venue__c

Standard Objects:

- Account
- Contact
- Case
- Task
- EmailMessage
- User

Automations:

- Record-Triggered Flows
- Auto-Launched Flows
- Approval Processes
- Scheduled Flows

Apex:

- Apex Classes for event-management operations
- Batch Apex for periodic processing

UI:

- Lightning App Page
- Dynamic Forms
- Lightning Record Pages
- Experience Cloud

Agentforce AI & Prompt Builder:

Fetch event data using Event ID and respond to event-related queries.

Email Services:

- Custom Email Alerts
- Email Templates

Security:

- Profiles
- Permission Sets
- Role Hierarchies
- Field-Level Security
- Sharing Rules

Designing Data Model and Security Model

Data Model

Event__c

Stores Event ID, Event Name, Event Date, Status, Capacity, and Event Description. It is linked with Organizer__c, Venue__c, and Registration__c.

Organizer__c

Stores organizer name, contact number, email, and organization information.

Attendee__c

Stores attendee name, contact number, email, and participant information.

Registration__c

Stores event registration information, registration date, registration status, and attendance status.

Venue__c

Stores venue name, location, capacity, and facilities.

Security Model

Role Hierarchy

Administrator → Event Manager → Event Staff → Participant

Admin Profile

Full access to objects and records.

Event Staff Profile

Read/Edit access to Event, Registration, Attendee, Organizer, and Venue records.

Participant Profile

Read-only access to their own event and registration details through Experience Cloud.

Support Staff Profile

Access to Cases and Event/Registration details for issue resolution.

Permission Sets

Provide additional privileges such as event-status updates, registration management, and Agent Console access.

Record-Level Security

Sharing Rules control access to Event__c and Registration__c.

Field-Level Security

Sensitive attendee contact information and organizer information are restricted to appropriate profiles.

Phase 2: Salesforce Development – Backend & Configurations

This phase focuses on implementing the core EventForce Management System logic. Custom objects are created for Event, Organizer, Attendee, Registration, and

Venue with relationships. Flows and approval processes are used to automate event status updates, registration activities, and participant notifications.

Milestone 1 – Salesforce Account

Salesforce is a customer success platform designed to support sales, service, marketing, analytics, and customer engagement.

Milestone 2 – Objects Creation

Event Object

Stores event name, event date, venue, capacity, description, and status.

Organizer Object

Stores organizer contact and organization information.

Attendee Object

Stores participant information.

Registration Object

Stores registration and attendance information.

Venue Object

Stores event venue information.

Milestone 3 – Tabs

Create tabs for:

- Events
- Organizers
- Attendees
- Registrations
- Venues

Milestone 4 – Fields & Relationships

Event Object

Event Name, Event Date, Status, Capacity, Description, Organizer, and Venue.

Organizer Object

Organizer Name, Email, Phone, and Organization.

Attendee Object

Attendee Name, Email, Phone, and Participant Type.

Registration Object

Registration Date, Registration Status, Attendance Status, Event, and Attendee.

Venue Object

Venue Name, Location, Capacity, and Facilities.

Milestone 5 – Prompt Builder

Create a Prompt Template for retrieving event details.

Testing

Use the Agent preview and enter a request such as:

“Show details for event EV-12345”

to retrieve the corresponding Event record details.

Milestone 6 – Flows

Create an Auto-Launched Flow or Record-Triggered Flow for event and registration automation.

Example Flow logic:

1. Start when an Event record is created or updated.
2. Check whether the Event Status has changed.
3. Get related Registration records.
4. Update registration or attendance information when required.
5. Send notifications to registered participants.
6. Save the updated records.

Phase 3: UI/UX Development & Customization

This phase focuses on enabling organizers to manage event operations efficiently while maintaining secure access through permission sets and AI-driven Agentforce experiences.

Milestone 1 – Creating Permission Sets

Create permission sets for:

- Event Manager
- Event Staff
- Participant
- Support Staff

Milestone 2 – Assigning Permissions to the Event User

Assign the required object, field, and system permissions to the appropriate users.

Milestone 3 – Org Setup for Agentforce

Enable the Salesforce default agent from Salesforce Setup and turn on the required services.

Milestone 4 – Building the Agent

Create the Event Management Agent with topics and actions for:

- Event information
- Event schedules
- Registration status
- Venue information
- Available seats
- Event status

Phase 4: Data Migration, Testing & Security

This phase validates the functionality and reliability of the Event Management Agent. It verifies topic recognition, Flow execution, event-data retrieval, registration processing, and permission handling so that the AI-driven agent operates correctly within Salesforce.

Testing includes:

- Event creation
- Event updates
- Registration creation
- Registration status updates
- Event status changes
- Automated notifications
- Agentforce responses
- Prompt Builder data retrieval
- User permissions
- Experience Cloud access

Phase 5: Deployment, Documentation & Maintenance

The purpose of this phase is to validate the Event Management Agent within the Salesforce Developer Org. Testing focuses on topic recognition, Flow execution, data retrieval, permissions, and agent interaction.

The project is tested within the Developer Org to simulate live behavior before deployment. Production UAT, sandbox deployment, and enterprise CI/CD are

outside the scope of this academic/prototype implementation because the Developer Edition environment is a standalone learning environment.

Maintenance and Monitoring Include

- Monitoring Event, Attendee, Registration, Organizer, and Venue data.
- Detecting incorrect agent queries, failed Flows, and performance delays.
- Resolving configuration errors.
- Updating Flow logic.
- Refining agent topics.
- Improving prompts and conversational interactions.
- Monitoring scalability and reliability in future enterprise deployments.

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, Salesforce screenshots were captured to validate the implemented functionality.

The project verifies:

- Event creation
- Organizer assignment
- Venue assignment
- Attendee creation
- Registration creation
- Registration status updates
- Event status updates
- Automation
- Agent interactions

- Data retrieval
- Permissions
- Dashboard-related functionality

The source document also notes that testing was used to verify correct functionality and performance.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralizes event and participant information in Salesforce.
- Improves visibility into event planning and registration.
- Automates key event-management processes using Salesforce Flow.
- Agentforce supports conversational participant assistance and event queries.
- Prompt Builder enables AI-powered responses and self-service.
- Reduces manual effort for participants, support teams, and administrators.
- Provides timely event notifications and updates.
- Experience Cloud can provide convenient participant access to event information.
- Supports integration with external event-management systems.
- Can scale for multiple events, venues, and growing participant operations.

Disadvantages

- Requires accurate and up-to-date event and participant data.
- Depends on Salesforce configuration and automation.
- Agentforce responses depend on properly configured prompts and available data.

- External event-management integrations can increase implementation complexity.
- Flow and automation logic require maintenance when business processes change.
- Users need appropriate Salesforce or Experience Cloud access.
- Initial implementation and configuration can require significant effort.
- Expansion to additional event types or integrations may require further customization.

9. CONCLUSION

EventForce Management System demonstrates how Salesforce can be used to develop a centralized event management solution. The project combines custom objects, relationships, Flows, Apex, Email Alerts, Experience Cloud, Agentforce AI, Prompt Builder, security controls, and reporting to support event creation, participant registration, scheduling, event-status updates, and communication.

The integration of Agentforce AI with Prompt Builder and Salesforce Flow provides a conversational approach to event management. It allows participants and organizers to obtain event information without depending entirely on manual support.

Overall, the project provides a practical foundation for a scalable and user-friendly event management solution.

10. FUTURE SCOPE

- Extend Agentforce capabilities for additional event-management queries.
- Enhance real-time event and attendance tracking.

- Expand participant self-service through Experience Cloud.
- Add advanced event-performance analytics.
- Integrate additional external event-management systems.
- Improve AI prompts and conversational experiences.
- Introduce enterprise-level sandbox and CI/CD deployment.
- Expand automation for additional event operations.
- Add QR-code-based event check-in.
- Introduce automated certificates for participants.
- Add event feedback and rating management.
- Implement advanced attendee analytics and engagement tracking.

Future enhancements can also include continuous monitoring, prompt refinement, Flow improvement, and increased AI-agent scalability.

11. APPENDIX

❖ **Custom Objects:** Event, Organizer, Attendee, Registration, Venue

❖ **Standard Objects:** Account, Contact, Case, Task, EmailMessage, User

❖ **Automation:** Flows, Approval Processes, Email Alerts

❖ **Apex:** Apex Classes, Batch Apex

❖ **AI:** Agentforce AI, Prompt Builder

❖ **UI:** Lightning App, Lightning Record Pages, Experience Cloud

❖ **Security:** Profiles, Roles, Permission Sets, Sharing Rules, Field-Level Security

❖ **Analytics:** Reports & Dashboards

❖ **Testing:** Flow, Agentforce, data retrieval, registration, security, and permission validation